MicroSense AI-Cam Report

AI-assisted optical monitoring report for preliminary screening of microplastic-like particle candidates in water samples.

1. Sample Summary

Report Generated	25 May 2026, 01:27 PM
Sample ID	12
Sample Source	Tap Water
Chamber Volume	50 ml
File Type	image
Monitoring Risk Level	**Low**

2. Detection and Monitoring Results

Detected Candidates	2
Estimated Particles / Litre	40
MSMI Score	17
MPI Score	17
Confidence Score	55
Particle Size Category	>500 µm estimated (prototype)
Average Particle Area	7648.50
Average Brightness	25.52

3. Hybrid Validation Results

Raw Detection Count	4
Accepted Candidate Count	2
Rejected Candidate Count	2
Hybrid Validation Score	43.09
Filter Summary	2 of 4 particle candidates were rejected. Main rejection reasons: weak contrast (2), invalic

4. Image Quality Assessment

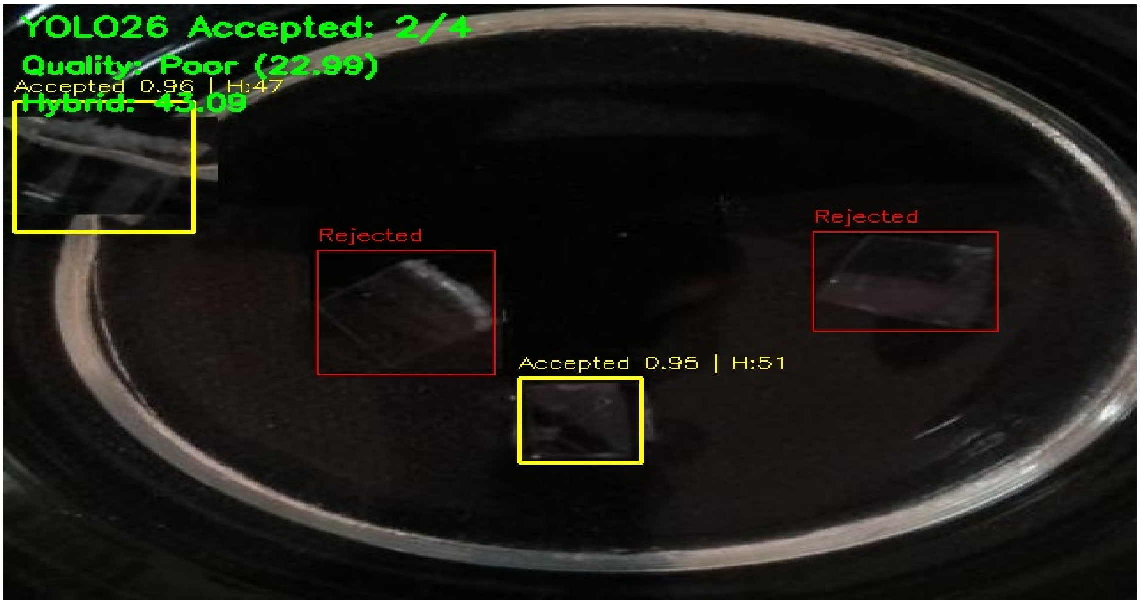
Image Quality Status	Poor
Image Quality Score	22.99
Focus Score	16
Brightness Score	16.10
Contrast Score	28.81
Overexposed Percent	0.01
Underexposed Percent	85.24
Quality Warning	Image may be blurry. Image is too dark. High underexposure detected. Low contrast imag

5. Risk Interpretation

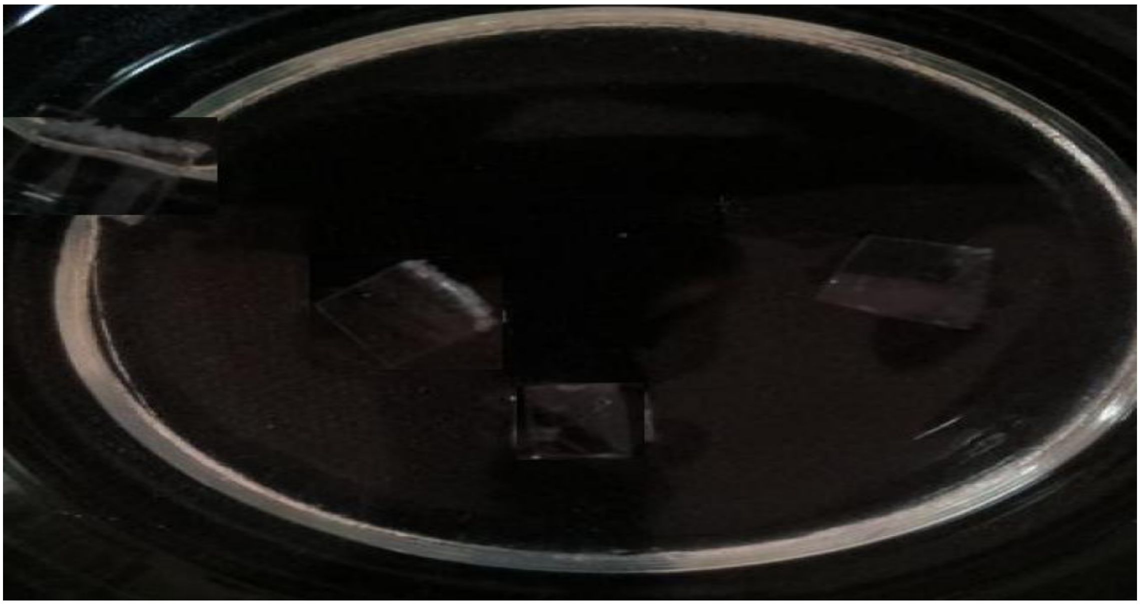
Source Risk Factor	1
Concentration Score	25
Size Score	20
Concentration-Only Risk Level	Low
Risk Explanation	MSMI is calculated using estimated particle concentration, detected count, average partic
Recommendation	Low monitoring concern. Continue normal monitoring and retest periodically.
Notes	Analyzed using YOLO26n main detector

6. Image Evidence

Processed Output Image



Original Uploaded Image



7. Scientific Scope and Disclaimer

This report is generated by MicroSense AI-Cam, a prototype-level AI-assisted optical monitoring system. The reported objects represent microplastic-like particle candidates identified from image-based visual features. The system does not chemically confirm polymer composition and does not replace certified laboratory techniques such as FTIR spectroscopy, Raman spectroscopy, or advanced microscopic analysis. The generated values, including estimated concentration, MSMI score, confidence score, and monitoring risk level, should be interpreted only as preliminary screening indicators and not as regulatory-grade or laboratory-certified measurements.

8. System Pipeline

Water sample image acquisition → YOLO26n-based candidate detection → OpenCV-based fallback support where required → hybrid visual validation → image quality assessment → monitoring score calculation → database storage → dashboard and report generation.